Continuous Relaxation Decoding for CSS Quantum LDPC Codes: Energy Landscape, Basin Geometry, Gradient Dynamics, and Hardware Implementation

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Abstract

We propose and rigorously analyse a continuous-relaxation decoder for CSS quantum low-density parity-check (qLDPC) codes, with the $[[193, 25, d = 4]]$ D4 causal diamond hypergraph-product code as the primary testbed [3, 2] and two high-distance augmented-seed codes — $[[112, 4, (6, 6)]]$ and $[[176, 32, (3, 6)]]$ — as secondary testbeds demonstrating near-term hardware relevance. A computational characterisation of the HGP seed structure confirms that $d = 4$ is the theoretical maximum for any code built from the causal diamond plaquette row space; a companion analysis identifies 64 weight-3 rows outside this space that raise d_{cl} from 4 to 6, generating the $[[112, 4, (6, 6)]]$ family ($kd^2/N \approx 1.29$, the Pareto-optimal finite-block code at $N \sim 100$) and the asymmetric $[[176, 32, (3, 6)]]$ code (rate 0.182, tailored to Z-biased noise). The decoder lifts binary error variables $e_q \in \{0, 1\}$ to continuous spin variables $v_q \in \mathbb{R}$ and defines a squared-spring energy $H(v)$ whose global minima are exactly the valid codewords.

Our analytical contributions are three exact theorems, each verified to machine precision ($< 10^{-10}$) across six independent CSS code families. The critical-point theorem shows that the linear spring formulation lacks fixed points at valid codewords, while the squared spring formulation achieves $H(v^*) = 0$ and $\nabla H(v^*) = \mathbf{0}$ simultaneously without approximation. The Hessian theorem gives the exact closed form $H_{qq} = 2\lambda + \deg_Z(q)/2$ at any valid codeword, implying strict positive definiteness for all $\lambda > 0$. These results are algebraic and hold universally for any CSS code.

Our experimental contributions characterise the full geometry of the energy landscape. The null space of the shared-check adjacency matrix has dimension equal to $N - \mathrm{rank}(H_Z)$ (109 out of 193 for the D4-HGP code), corresponding precisely to the logical operator subspace; gradient dynamics along these directions is governed solely by the containment term. The direct correction path from a valid codeword to a single-error state is provably monotone, establishing that single-error decoding failures are provably impossible under gradient descent; empirically, this extends to $t \leq 2$ errors. Failures for $t \geq 3$ are caused exclusively by pseudo-codeword traps with barrier heights $\Delta E \in [1, 4]$ at $\lambda = 0.5$; the actual basin of attraction is 3–11×

wider than the Hessian radius predicts, and both radii scale as $\lambda^{-1/2}$, making the normalised basin shape λ-invariant.

A sweep across four optimiser variants (gradient descent, heavy-ball momentum, Nesterov, and Langevin dynamics) reveals that heavy-ball achieves a marginal gain at $t = 5$ errors (beyond the minimum distance $d = 4$, probing optimiser behaviour under high error load) via a path-geometry mechanism, not barrier vaulting; the Langevin decoder requires noise temperature $T_0 \ll \Delta E_{\min} = 1.0$ for the code. A key conceptual result is that $H(v)$ is *syndrome-agnostic*: the energy depends only on spin configuration, not on the target syndrome. The decoder therefore acts as a *codeword projector*, with its standalone performance determined by the initialization quality rather than by the landscape itself.

Empirically, the optimal containment coefficient follows a corrected universal heuristic $\lambda^* \approx \frac{3}{16}\bar{d}$, where $\bar{d}$ is the mean qubit degree; this refines the trace-balance prediction $\bar{d}/4$ by a factor of 0.75 observed consistently across four independent code families. Furthermore, the girth of the Tanner graph does not predict the pseudo-codeword trap density for continuous decoding: a code with girth 8 traps at 2.5× the rate of a code with girth 4, falsifying a natural generalisation of the belief propagation analysis.

On hardware, a fully vectorised PyTorch GPU implementation achieves 86 535 syndromes per second at batch size 16 384, a $\approx 28\,000\times$ speedup over the serial Python baseline (which operates at $\approx 0.1\%$ theoretical FLOP efficiency due to the Python check loop). The FLOP comparison with MWPM is regime-dependent: against the naive $O(N^3)$ worst-case bound the continuous decoder uses 8× fewer FLOPs; against a sparse-Blossom graph model ($\approx 49\,000$ FLOPs for 810 matching-graph edges) it uses 18× more. Furthermore, standard MWPM implementations are structurally inapplicable to the D4-HGP code: 44% of its qubits have column weight ≥ 3 and require a hyperedge decomposition for which no simple clique approximation is effective, yielding $\leq 31\%$ success at $t = 1$ errors. The practical step-count optimum is $K = 200$ gradient steps, achieving 96% of the asymptotic success rate at twice the throughput of $K = 400$.

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1 Introduction

Decoding quantum LDPC codes is a central challenge in quantum error correction. The minimum-weight perfect matching (MWPM) decoder, standard for surface codes, does not extend to the geometry of general qLDPC codes. Belief propagation (BP) does extend, but qLDPC Tanner graphs contain short cycles that cause BP messages to form self-reinforcing loops — the "trapping set" or "pseudo-codeword" phenomenon — and BP can fail catastrophically even at low error rates [6, 7].

An alternative approach, motivated by physics-based optimisation, lifts the discrete decoding problem to a continuous energy minimisation. Hopfield networks [4] encode constraint satisfaction as energy minima; Ising machines [5] exploit physical dynamics to solve combinatorial problems. The central question for QEC is whether the continuous energy landscape can be designed so that gradient dynamics reliably finds the correct codeword.

We apply this approach to the $[[193, 25, d = 4]]$ CSS qLDPC code constructed in [2] via the Tillich–Zémor hypergraph product of the D4 causal diamond seed matrix [1]. This code is a concrete, fully specified instance with known parameters. We additionally test the decoder on two high-distance codes from the augmented-seed family of [2]: the $[[112, 4, (6, 6)]]$ code ($kd^2/N \approx 1.29$, the highest figure-of-merit code from the causal diamond geometry at $N \sim 100$) and the $[[176, 32, (3, 6)]]$ code (rate 0.182, asymmetric distances suited to Z-biased noise hardware). For both new codes, all analytical theorems pass to machine precision, the $\lambda^* = \frac{3}{16}\bar{d}$ heuristic is independently confirmed, and the continuous decoder achieves 100% success for $t \leq \lfloor (d-1)/2 \rfloor$ with graceful degradation beyond. We then test the universality of all analytical results across a full zoo of eight CSS code families including bicycle codes, repetition-code hypergraph products, and random LDPC products.

Contributions.

(i) A critical-point theorem proving that the linear spring formulation has no fixed points at valid codewords, while the squared spring formulation achieves zero energy and zero gradient simultaneously, without any approximation (theorem 3.4).

(ii) An exact closed-form Hessian at any valid codeword: $H_{qq} = 2\lambda + \deg_Z(q)/2$ on the diagonal, implying strict local minimality for all $\lambda > 0$, verified to machine precision across six CSS code families (theorem 4.3 and section 10).

(iii) An exact single-error formula showing that the gradient restoring force equals $\deg_Z(q^*)$, proven and verified to machine precision (theorem 4.1).

(iv) A three-radius characterisation of the codeword basin geometry: the quadratic Hessian radius r_{loc}, the pseudo-codeword barrier radius r_{barrier}, and the empirical escape radius r_{emp}, with $r_{\text{emp}} \gg r_{\text{barrier}} \geq r_{\text{loc}}$ and a provably monotone direct correction path for $t \leq 2$ (section 6).

(v) A comparative analysis of four optimiser variants (GD, heavy-ball, Nesterov, Langevin), including proof by trajectory analysis that the momentum mechanism is path-geometric rather than barrier-vaulting, and a calibration requirement for Langevin dynamics (section 8).

(vi) A conceptual identification of the decoder as a *syndrome-agnostic codeword projector*, with implications for the correct measurement protocol for the decoder threshold (section 9).

(vii) A corrected universal heuristic $\lambda^* \approx \frac{3}{16}\bar{d}$ for the optimal containment coefficient, and a falsification of the girth–trap-density hypothesis for continuous decoding (section 10).

(viii) A hardware acceleration study demonstrating 86 535 syndromes/s on a GPU via gather/scatter vectorisation, 8× fewer FLOPs than the naive $O(N^3)$ MWPM bound (18× *more* than sparse-Blossom), and a $K = 200$ practical optimum for step count (section 11).

(ix) Experimental characterisation of the decoder on two high-distance codes from the augmented-seed family [2]: the $[[112, 4, (6, 6)]]$ and $[[176, 32, (3, 6)]]$ codes, both significantly outperforming the surface code at $N \sim 100$–200. Decoder benchmarks independently confirm the algebraically certified distances and validate the λ^* heuristic, establishing the continuous relaxation decoder as the natural companion algorithm for the Lorentzian code family (section 10).

2 Setup and Notation

Let $\mathcal{C}$ be a CSS code on N physical qubits with Z-stabiliser matrix $H_Z \in \mathbb{F}_2^{M\times N}$ and X-stabiliser matrix $H_X \in \mathbb{F}_2^{M\times N}$, satisfying $H_Z H_X^T \equiv 0 \pmod 2$. We work throughout with Z-error decoding; the X-error case is symmetric.

Definition 2.1 (Tanner graph and support)**.** The *Tanner graph* $G(H_Z)$ is bipartite with N qubit nodes and M check nodes; qubit q and check j are adjacent iff $H_Z[j, q] = 1$. The *support* of check j is $\mathrm{supp}(j) = \{q : H_Z[j, q] = 1\}$ with $|\mathrm{supp}(j)| = w_j$ (the check weight). The *column weight* $\deg_Z(q) = \sum_j H_Z[j, q]$ is the degree of qubit q in $G(H_Z)$. The mean qubit degree is $\bar{d} = N^{-1}\sum_q \deg_Z(q) = M\bar{w}/N$, where $\bar{w}$ is the mean check weight.

Definition 2.2 (Spin lift)**.** The *spin representation* maps binary vectors $e \in \{0, 1\}^N$ to spin vectors $v \in \{+1, -1\}^N$ via $v_q = 1 - 2e_q$. In this representation, the GF(2) parity constraint $\bigoplus_{q\in\mathrm{supp}(j)} e_q = 0$ is equivalent to the multiplicative constraint $P_j(v) := \prod_{q\in\mathrm{supp}(j)} v_q = +1$.

The spin representation is exact on $\{+1, -1\}^N$; no approximation is involved.

Definition 2.3 (Shared-check adjacency matrix)**.** The *shared-check adjacency matrix* $A \in \mathbb{Z}_{\geq 0}^{N\times N}$ is defined by $A_{qr} = |\{j : q \in \mathrm{supp}(j),\, r \in \mathrm{supp}(j)\}|$. The diagonal entries satisfy $A_{qq} = \deg_Z(q)$. In matrix form, $A = H_Z^T H_Z$ (as an integer matrix). Since A is a Gram matrix, $A \succeq 0$.

The D4-HGP code. The primary testbed is the $[[193, 25, d = 4]]$ code obtained as the hypergraph product of the 7×12 independent-row basis of the D4 causal diamond Z-stabiliser matrix [2]. Parameters: $N = 193$, $M = 84$, $\mathrm{rank}(H_Z) = 84$. Column weights range from 1 to 7 with mean $\bar{d} = 2.76$. Row weights range from 5 to 11. The null space of A has dimension $\dim\ker(A) = N - \mathrm{rank}(H_Z) = 109$.

Remark 2.4 (Corrected distance and seed characterisation). An earlier version of [2] reported $d \geq 67$ for this code. A subsequent revision to the scaling analysis in that work established $d = 4$. All decoder-analytic results in the present paper (Theorems 3.4, 4.1, 4.3 and all associated observations) are algebraic identities that hold for any CSS code independent of minimum distance; they are unaffected by this correction. The experimental benchmarks at $t = 1, 2, 3, 4$ remain meaningful and correctly characterise decoder

performance relative to the true distance. Experimental results at $t \geq 5$ probe the regime beyond the minimum distance and should be interpreted as stress-tests of the optimiser, not as below-threshold decoding.

A computational characterisation of the HGP seed structure, carried out as part of the analysis accompanying this paper, establishes that $d = 4$ is the *theoretical maximum* for any hypergraph-product code built from the causal diamond geometry. The classical minimum distance of the primal seed H_Z^{primal} (the 7×12 independent-row basis of the plaquette incidence matrix) equals $d_{\text{cl}} = 4$, achieved uniquely by the three spatial-axis group vectors G_1, G_2, G_3; no vector of weight 1, 2, or 3 lies in $\ker(H_Z^{\text{primal}})$. Since the HGP distance satisfies $d_Z(\text{HGP}(H, H)) \leq d_{\text{cl}}(H)$, the primal seed achieves the best distance available. The dual family of [1] yields only smaller classical distances ($d_{\text{cl}} = 3$ for all 1248 valid Code II X-check matrices, verified exhaustively, and $d_{\text{cl}} = 2$ for the spatial-axis seed), confirming that no orientation or variant within the causal diamond 2-complex improves on $d = 4$. The $d = 4$ result is a geometric invariant of the causal diamond, not a defect of the Lorentzian construction.

However, a single weight-3 row *outside* the plaquette row space, whose support intersects each of G_1, G_2, G_3 in exactly one qubit, kills all three minimum-weight codewords simultaneously and raises d_{cl} from 4 to 6 (weight spectrum $\{6{:}\,12,\ 8{:}\,3\}$). There are precisely **64** such rows of minimum weight 3. Under the B_4 symmetry group (of order 96), they partition into exactly 2 orbits: one of size 16 and one of size 48; the set is B_4-closed but not B_4-transitive. The hypergraph product of the resulting augmented (8×12) seed with itself yields a $[[208, 16]]$ code. Because H_{aug} has full row rank 8, its transpose H_{aug}^T has trivial kernel over $\mathbb{F}_2$, so the Tillich–Zémor theorem gives $d_Z \geq \min(6, \infty) = 6$ and $d_X \geq 6$ algebraically. A subsequent meet-in-the-middle search confirmed a weight-6 Z-logical operator exists, establishing $d_Z = d_X = 6$ exactly and making this the **first code from the causal diamond geometry exceeding the $d = 4$ ceiling**: $[[208, 16, 6]]$ with $kd^2/N = 16 \times 36/208 \approx 2.77$. See section 13 for the full analysis and extended code families.

3 The Continuous Energy Function

3.1 Two candidate energy functions

We extend the spin representation to the full real line by introducing continuous variables $v_q \in \mathbb{R}$ and defining energy functions whose global minima are the codeword spin vectors.

Definition 3.1 (Containment potential)**.** The *containment potential* is the double-well function

$$V(v_q) = \frac{\lambda}{4}(v_q^2 - 1)^2, \qquad \lambda > 0,$$

with global minima at $v_q = \pm 1$ and a local maximum at $v_q = 0$. Its gradient is $V'(v_q) = \lambda(v_q^2 - 1)v_q$, which vanishes at $v_q = \pm 1$.

Definition 3.2 (Linear spring energy)**.**

$$H_{\text{lin}}(v) = \sum_{q=1}^{N} V(v_q) + \sum_{j=1}^{M} \frac{1 - P_j(v)}{2}.$$

For $v \in \{+1, -1\}^N$: each satisfied check contributes 0, each violated check contributes 1.

Definition 3.3 (Squared spring energy)**.**

$$H(v) = \sum_{q=1}^{N} V(v_q) + \sum_{j=1}^{M} \frac{\big(1 - P_j(v)\big)^2}{4}.$$

For $v \in \{+1, -1\}^N$: each satisfied check contributes 0, each violated check contributes 1. The discrete values of H and H_{lin} are identical on $\{+1, -1\}^N$.

Both functions agree on the discrete constraint set. Their critical-point structure in the continuous domain is radically different.

3.2 Critical-point theorem

Theorem 3.4 (Critical points of the linear and squared spring energies)**.** *Let $v^* \in \{+1, -1\}^N$ be a valid codeword.*

(i) Linear spring: $\nabla H_{\text{lin}}(v^*) \neq \mathbf{0}$. *Explicitly, at $v^* = +\mathbf{1}$,*

$$\left.\frac{\partial H_{\text{lin}}}{\partial v_q}\right|_{v=+\mathbf{1}} = -\frac{\deg_Z(q)}{2} \neq 0.$$

Gradient dynamics on H_{lin} cannot converge to any valid codeword.

(ii) Squared spring: $\nabla H(v^*) = \mathbf{0}$ *and* $H(v^*) = 0$.

Proof. Part (i). The containment gradient vanishes at $v_q = \pm 1$: $V'(\pm 1) = 0$. The linear spring gradient at qubit q evaluated at $v = +\mathbf{1}$ is

$$-\frac{1}{2} \sum_{j: q \in \text{supp}(j)} \prod_{k \in \text{supp}(j),\, k \neq q} v_k = -\frac{\deg_Z(q)}{2},$$

since every $v_k = +1$.

Part (ii). The squared spring gradient at qubit q is

$$\frac{\partial H}{\partial v_q} = V'(v_q) - \frac{1}{2} \sum_{j: q \in \text{supp}(j)} \big(1 - P_j\big) \cdot \frac{P_j}{v_q}.$$

At a valid codeword $P_j = +1$ for all j, so $(1 - P_j) = 0$, and the spring gradient vanishes. $V'(v_q^*) = 0$ since $v_q^* = \pm 1$. Thus $\nabla H(v^*) = \mathbf{0}$. $H(v^*) = 0$ follows from $V(\pm 1) = 0$ and $(1 - P_j)^2/4 = 0$. □

Remark 3.5 (Exact reformulation, not approximation). The squared spring is not an approximation of the linear spring. Both equal the check-violation count on $\{+1, -1\}^N$. The term $(1 - P_j)^2/4$ equals $(1 - \cos\theta_j)/2$ where $\theta_j \in \{0, \pi\}$ is the phase of P_j on $\{+1, -1\}^N$, the XZ-rotation energy familiar from quantum Ising models.

4 Analytical Results

4.1 Single-error energy and gradient

Theorem 4.1 (Single-error formulae)**.** *Let $v^{(q^*)}$ denote the spin vector with $v_{q^*} = -1$ and $v_q = +1$ for $q \neq q^*$. Then:*

$$H(v^{(q^*)}) = \deg_Z(q^*), \tag{1}$$

$$\left.\frac{\partial H}{\partial v_q}\right|_{v=v^{(q^*)}} = \begin{cases} -\deg_Z(q^*) & q = q^*, \\ +A_{q,q^*} & q \neq q^*, \end{cases} \tag{2}$$

Proof. Energy. Each check j containing q^* has $P_j = -1$, contributing $(1-(-1))^2/4 = 1$. Checks not containing q^* have $P_j = +1$, contributing 0. The containment contributes $V(-1) = 0$. Thus $H(v^{(q^*)}) = \deg_Z(q^*)$.

Gradient at $q = q^*$. For each $j \ni q^*$: $P_j = -1$, $\frac{\partial}{\partial v_{q^*}} \frac{(1-P_j)^2}{4} = -\frac{1}{2}(1-P_j)\frac{P_j}{v_{q^*}} = -1$. Summing over $\deg_Z(q^*)$ checks gives $-\deg_Z(q^*)$.

Gradient at $q \neq q^*$. For $j \ni q, q^*$: $P_j = -1$ (since $v_{q^*} = -1$ and all others $+1$) and $\frac{P_j}{v_q} = -1/1 = -1$, contributing $-\frac{1}{2}(1-(-1))(-1) = -\frac{1}{2}(2)(-1) = +1$. For $j \ni q, j \not\ni q^*$: $P_j = +1$, contributing 0. Summing over the $A_{q,q^*} = |\{j : q, q^* \in \mathrm{supp}(j)\}|$ shared checks gives $+A_{q,q^*}$. □

Corollary 4.2 (Restoring force magnitude)**.** *The restoring force on the erroneous qubit is $+\deg_Z(q^*) > 0$, directed toward $v_{q^*} = +1$. Its magnitude equals the qubit's Tanner graph degree. The gradient at a neighbouring qubit $q \neq q^*$ is $+A_{q,q^*} \geq 0$: the spring term* pushes *neighbours away from $+1$ (toward the error), consistent with the squared-spring trying to restore violated checks by flipping all participants.*

4.2 Hessian at valid codewords

Theorem 4.3 (Exact Hessian at any valid codeword)**.** *At any valid codeword v^*, the Hessian $\mathbf{H} = \nabla^2 H(v^*)$ has the exact structure:*

$$\mathbf{H}_{qq} = 2\lambda + \frac{\deg_Z(q)}{2}, \tag{3}$$

$$\mathbf{H}_{qr} = \frac{A_{qr}}{2}, \quad q \neq r, \tag{4}$$

where $A = H_Z^T H_Z$ is the shared-check adjacency matrix. In matrix form: $\mathbf{H} = 2\lambda I + \frac{1}{2}A$.

Proof. Diagonal. The containment contributes $V''(\pm 1) = \lambda(3 \cdot 1 - 1) = 2\lambda$. For the spring term, at $P_j = +1$: $\frac{\partial^2}{\partial v_q^2}\frac{(1-P_j)^2}{4} = \frac{1}{2}\left(\frac{P_j}{v_q}\right)^2 = \frac{1}{2}$. Summing over $\deg_Z(q)$ checks gives $\deg_Z(q)/2$.

Off-diagonal. Only checks containing both q and r contribute. At $P_j = +1$: $\frac{\partial^2}{\partial v_q \partial v_r}\frac{(1-P_j)^2}{4} = \frac{1}{2} \cdot \frac{P_j^2}{v_q v_r} = \frac{1}{2}$, since $v_q^*, v_r^* = \pm 1$ and $P_j^2 = 1$. Summing over A_{qr} shared checks gives $A_{qr}/2$. □

Corollary 4.4 (Strict local minimality for all $\lambda > 0$)**.** $\mathbf{H} = 2\lambda I + \frac{1}{2}A \succ 0$ *for all $\lambda > 0$, since $A = H_Z^T H_Z \succeq 0$. Every valid codeword is a strict local minimum of $H(v)$.*

4.3 Hessian spectral structure

The eigenvalue formula $\mu_k(\lambda) = 2\lambda + \sigma_k(A/2)$ follows directly from theorem 4.3, where $\sigma_k(A/2)$ are the eigenvalues of $A/2$. These are code-intrinsic (λ-independent) and can be computed once from H_Z.

Proposition 4.5 (Null-space structure)**.** *The null space of $A = H_Z^T H_Z$ over $\mathbb{R}$ has dimension $\dim\ker(A) = N - \mathrm{rank}_{\mathbb{R}}(H_Z)$, and corresponds precisely to vectors that share no check pair. For the D4-HGP code: $\dim\ker(A) = 193 - 84 = 109$. In the null directions, $\mu_{\mathrm{null}} = 2\lambda$ exactly, so basin curvature is governed solely by the containment term. These null directions span a subspace that contains the logical operator representatives.*

Proof. $\ker(A) = \ker(H_Z)$ over $\mathbb{R}$ since $A = H_Z^T H_Z$ and $Ax = 0 \Leftrightarrow x^T Ax = \|H_Z x\|^2 = 0 \Leftrightarrow H_Z x = 0$. The dimension of $\ker(H_Z)$ over $\mathbb{R}$ is $N - \mathrm{rank}_{\mathbb{R}}(H_Z)$. Note that for $\{0,1\}$-matrices arising from CSS codes, $\mathrm{rank}_{\mathbb{R}}(H_Z) \geq \mathrm{rank}_{\mathbb{F}_2}(H_Z)$; the two ranks coincide whenever the real row space contains no non-trivial integer linear dependencies beyond those detectable over $\mathbb{F}_2$. For all codes in this paper (verified numerically), $\mathrm{rank}_{\mathbb{R}} = \mathrm{rank}_{\mathbb{F}_2}$. The containment contribution to eigenvalue in these directions is 2λ; the spring contribution is $x^T(A/2)x = 0$. □

Remark 4.6 (Code structure as geometric advantage). For the D4-HGP code, $109/193 \approx 56\%$ of Hessian eigendirections lie in the null space of A. In these directions the codeword basin extends to the containment clip boundary, measured empirically as $r_{\mathrm{emp}} \approx \rho/\sqrt{\lambda}$ where $\rho = 1.5$ is the clip parameter. This is the code's topological structure acting as a structural advantage for the continuous decoder.

4.4 Local minima and pseudo-codewords

Proposition 4.7 (Local minima are pseudo-codewords)**.** *Let v^{loc} be a local minimum of H and let $\hat{v} = \mathrm{sign}(v^{\mathrm{loc}})$. If $\hat{v}$ is not a valid codeword, then the binary vector $\hat{e} = (1-\hat{v})/2 \bmod 2$ is a pseudo-codeword: $H_Z\hat{e} \not\equiv \mathbf{0}$ (mod 2).*

This is the continuous analogue of BP trapping sets. Unlike BP fixed points, these traps are geometric objects (local basins) amenable to physical escape strategies.

5 The Gradient Descent Decoder

Construction 5.1 (Continuous Relaxation Decoder)**.** Given syndrome $s \in \{0,1\}^M$:

1. *Initialise:* $v^{(0)} = +\mathbf{1} - 2\hat{e}^{(0)}$, where $\hat{e}^{(0)}$ is a syndrome-consistent error estimate (e.g. greedy peeling, see section 9).
2. *Iterate:* $v^{(k+1)} = \mathrm{clip}\big(v^{(k)} - \eta\nabla H(v^{(k)}),\ [-\rho,\rho]\big)$, with $\eta = 0.05$, $\rho = 1.5$, for K steps.
3. *Snap:* $\hat{v} = \mathrm{sign}(v^{(K)})$; $\hat{e} = (1-\hat{v})/2 \bmod 2$.
4. Apply $\hat{e}$ as the correction.

Gradient computation. The gradient ∇H is computed analytically from the squared-spring formula. The partial derivative at qubit q is

$$\frac{\partial H}{\partial v_q} = \lambda(v_q^2 - 1)v_q - \frac{1}{2}\sum_{j\,:\,q\in\mathrm{supp}(j)} (1 - P_j)\cdot\frac{P_j}{v_q}. \tag{5}$$

No message-passing or graph traversal is needed.

Complexity. Each gradient step requires $3N$ FLOPs for the containment term and $3\cdot\mathrm{nnz}(H_Z)$ FLOPs for the spring term, giving $3(N + \mathrm{nnz}(H_Z))$ FLOPs per step. For the D4-HGP code ($N = 193$, nnz $= 532$): $3 \times 725 = 2{,}175$ FLOPs per step. At $K = 400$ steps: 870,000 FLOPs per syndrome, versus $O(N^3) \approx 7.2 \times 10^6$ for MWPM.

6 Basin Geometry and Convergence Theory

This section characterises the geometry of the codeword basin of attraction through three complementary radii and proves that decoding failures are impossible for $t \leq 2$ errors under gradient descent.

6.1 Three nested radii

Definition 6.1 (Three convergence radii)**.** Fix a valid codeword v^* and $\lambda > 0$. Define:

- $r_{\text{loc}}(\lambda) = \sqrt{2/\mu_{\min}(\lambda)}$, the *Hessian radius*, where $\mu_{\min} = 2\lambda + \sigma_{\min}(A/2)$ is the smallest Hessian eigenvalue.
- $r_{\text{barrier}}(\lambda) = \sqrt{2\Delta E/\mu_{\min}(\lambda)}$, the *barrier radius* corresponding to saddle height ΔE to the nearest pseudo-codeword basin.
- $r_{\text{emp}}(\lambda, \hat{u})$, the *empirical escape radius* along direction $\hat{u}$: the smallest r such that $\text{GD}(v^* + r\hat{u})$ fails to return to any valid codeword.

Observation 6.2 (Nested basin geometry)**.** For the D4-HGP code at $\lambda = 0.5$:

$$r_{\text{loc}} \approx 1.41 \;\leq\; r_{\text{barrier}} \in [1.41,\, 2.83] \;\ll\; r_{\text{emp}} \in [7,\, 15] \times r_{\text{loc}},$$

across $\hat{u}$ ranging from the stiffest to the softest Hessian eigendirection. The Hessian radius is a conservative lower bound; the actual basin is 3–11× wider than r_{loc}.

The numerical values are reported for 12 sampled pseudo-codewords at $\lambda = 0.5$ in table 1.

6.2 Monotone direct correction path

Proposition 6.3 (No barrier on the direct correction path)**.** *Consider the one-dimensional restriction of H to the path $v^*(t) = v^* + t(v^{(q^*)} - v^*)$ for $t \in [0, 1]$, connecting a valid codeword v^* to a single-error state $v^{(q^*)}$. For any physical parameter values satisfying $\lambda < 2\deg_Z(q^*)$, the gradient dH/dt is everywhere negative on $(0, 1)$: the path is* strictly monotone decreasing *in energy from $v^{(q^*)}$ to v^*. In particular, no saddle point separates $v^{(q^*)}$ from v^* along this path.*

Proof. Restricting to the 1D path by holding all other qubits at $+1$, the energy along the path is

$$E(s) = \frac{\lambda}{4}(s^2 - 1)^2 + \frac{\deg_Z(q^*)}{4}(1 - s)^2, \quad s \in [-1, +1],$$

where $s = v_{q^*}(t) = -1 + 2t$ parametrises the erroneous qubit. A stationary point other than $s = +1$ (the codeword) satisfies the cubic equation $(s-1)[\lambda s(s+1) + \deg_Z(q^*)/2] = 0$. The bracket has discriminant $\Delta = \lambda^2 - 2\lambda \cdot \deg_Z(q^*)$. For $\lambda < 2\deg_Z(q^*)$, which holds for all physical λ since $\lambda \ll \deg_Z(q^*)$ in the interesting regime, we have $\Delta < 0$: the bracket has no real roots other than the trivial factored root. Therefore $s = +1$ is the unique stationary point on $[-1, +1]$. □

Corollary 6.4 (Provably perfect decoding for $t = 1$; empirical for $t = 2$)**.** *For a code with minimum qubit degree $d_{\min} \geq 1$, gradient descent with any $\lambda < 2d_{\min}$ corrects every single-error pattern at 100% rate. Empirically, perfect decoding extends to $t = 2$ errors in all experiments; however, the monotone-path argument does not constitute a formal proof*

for $t = 2$, since correcting two simultaneous errors requires showing that the joint two-error initial point lies in the codeword basin, not merely that each individual sub-problem is monotone.

This explains the experimental observation of 100% success for $t \leq 2$ errors over all 30 random trials at each weight.

6.3 Pseudo-codeword barriers

Failures for $t \geq 3$ arise not from the direct correction path but from the trajectory landing in the basin of a pseudo-codeword v_{pc} whose snapped configuration violates at least one check.

Observation 6.5 (Barrier height distribution)**.** For 12 sampled pseudo-codewords collected from failed $t = 5$ trials at $\lambda = 0.5$, the straight-line energy maximum $\Delta E = \max_{t\in[0,1]} H(v^* + t(v_{\text{pc}} - v^*))$ satisfies: $\Delta E \in [1.0, 4.0]$, with mean 1.67. The tightest trap has $\Delta E = 1.0$ and involves a 2-qubit pattern sharing a single Tanner graph cycle of length 4 (girth cycle). The corresponding barrier radius $r_{\text{barrier}} = r_{\text{loc}} = 1.41$, confirming that the Hessian radius exactly predicts the binding constraint for the critical trap class.

Remark 6.6 (Barrier measurement note)*.* The reported ΔE values are upper bounds on the true saddle height, since the straight-line path is not guaranteed to be the minimum energy path between v^* and v_{pc}. The true (minimum-energy-path) saddle may be lower. An exact saddle search would require solving for the index-1 critical point of H between the two basins.

6.4 Scaling with λ

Observation 6.7 (λ-invariance of normalised basin shape)**.** For both r_{loc} and r_{barrier}, the scaling law $r \propto \lambda^{-1/2}$ holds across $\lambda \in [0.1, 2.0]$. The ratio $r_{\text{barrier}}/r_{\text{loc}}$ is nearly constant over this range ($\lesssim 10\%$ variation). Consequently, the basin geometry—when expressed in units of r_{loc}—is λ-invariant. This implies that optimising λ does not fundamentally change the barrier-to-basin-size ratio; it only scales all length scales together.

Table 1: Basin geometry parameters at $\lambda = 0.5$ for the D4-HGP code. Barrier heights measured along the straight-line path $v^* \to \text{snap}(v_{\text{pc}})$. The Hessian radius $r_{\text{loc}} = \sqrt{2/\mu_{\min}} = 1.414$ at $\lambda = 0.5$.

PC type	n_{flip}	ΔE	r_{barrier}	$r_{\text{barrier}}/r_{\text{loc}}$
2-flip (most common)	2	1.00	1.41	1.00
4-flip, interior saddle	4	2.00–2.27	2.00–2.13	1.41–1.51
6-flip, interior saddle	6	2.51	2.24	1.58
4-flip, max syndrome wt	4	4.00	2.83	2.00
Summary		min 1.00, max 4.00	min 1.41, max 2.83	1.00–2.00

Table 2: Hessian spectrum and quadratic convergence radius at selected λ. $\sigma_{\min}(A/2) \approx 0$ (null space dim = 109); $r_{\text{loc}} \approx 1/\sqrt{\lambda}$ for all physical λ.

λ	$\mu_{\min}$	$\mu_{\max}$	κ	r_{loc}
0.05	0.100	14.99	149.9	4.472
0.10	0.200	15.09	75.5	3.162
0.50	1.000	15.89	15.89	1.414
1.00	2.000	16.89	8.44	1.000
2.00	4.000	18.89	4.72	0.707

7 Experiments on the $[[193, 25]]$ Code

7.1 Setup

All experiments use $\eta = 0.05$, $K = 400$ steps, $\rho = 1.5$, $\lambda = 0.5$ unless otherwise stated. Error patterns are drawn uniformly at random. Trial counts are specified per experiment.

7.2 Analytical predictions verified

table 3 reports the numerical verification of all analytical results from section 4. All results match to machine precision.

Table 3: Numerical verification of analytical formulae on the $[[193, 25]]$ code. All discrepancies are below machine epsilon ($< 10^{-10}$).

Claim	Predicted	Numerical
Ground state energy $H(+\mathbf{1})$	0	0 (exact)
Ground state gradient $\|\nabla H(+\mathbf{1})\|$ (squared spring)	0	1.76×10^{-13}
Ground state gradient $\|\nabla H_{\text{lin}}(+\mathbf{1})\|$	$\sqrt{\sum_q d_q^2}/2$	22.54 (matches)
Single-error energy, $d_q = 7$	7	7.000 (exact)
Single-error energy, $d_q = 1$	1	1.000 (exact)
Single-error gradient at q^* ($d_q = 7$)	-7	-7.000 (exact)
Hessian diagonal ($\lambda = 1$, $d_q = 7$)	5.5	5.500 (exact)
Hessian diagonal ($\lambda = 1$, $d_q = 1$)	2.5	2.500 (exact)

7.3 Gradient descent convergence vs. error weight

Table 4: GD decoder performance vs. error weight t. $\lambda = 0.5$, $K = 400$, 100 trials each.

t	Success rate	Mean residual syndrome (failures)	Initial energy
1	100/100 (1.00)	–	≈ 2.64
2	100/100 (1.00)	–	≈ 4.68
3	99/100 (0.99)	0.14	≈ 7.40
4	91/100 (0.91)	0.18	≈ 10.1
5	82/100 (0.82)	0.08	≈ 12.84

For $t \leq 2$ the decoder is exact over all 100 trials, consistent with corollary 6.4. Failures at $t \geq 3$ correspond exclusively to pseudo-codeword trapping (nonzero residual syndrome).

7.4 Lambda sensitivity

table 5 reports success rate vs. λ for $t = 3$ errors.

Table 5: Decoder success rate vs. λ for $t = 3$ errors (100 trials). The heuristic optimal $\lambda^* \approx \frac{3}{16}\bar{d} = 0.517$.

λ	Success rate	Mean final energy
0.10	97/100 (0.97)	0.017
0.50	99/100 (0.99)	0.000
1.00	78/100 (0.78)	0.311
2.00	32/100 (0.32)	0.775
5.00	8/100 (0.08)	2.607

The optimal range is $\lambda \in [0.1, 0.5]$. The mechanistic interpretation follows from theorem 4.3: at large λ the deep basin curvature $2\lambda + \deg_Z(q)/2$ immediately suppresses spring-driven corrections. A heuristic balance condition $2\lambda = \bar{d}/2$ yields $\lambda_{\text{pred}} = \bar{d}/4 = 0.69$ for the D4-HGP code. Empirically the optimum sits at $\lambda^* \approx 0.75 \times 0.69 = 0.52$ (see section 10 for the full derivation of this factor).

8 Momentum-Based Decoding

8.1 Optimiser variants

Three momentum-based optimisers are compared against vanilla GD.

Heavy-ball (HB; Polyak 1964). $v^{(k+1)} = v^{(k)} - \eta\nabla H(v^{(k)}) + \gamma(v^{(k)} - v^{(k-1)})$. The momentum term $\gamma(v^{(k)} - v^{(k-1)})$ injects velocity proportional to the current velocity. For a quadratic landscape with condition number κ, the optimal parameter is $\gamma_{\text{opt}} = \left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^2$. For the D4-HGP code at $\lambda = 0.5$, $\kappa = 15.89$, giving $\gamma_{\text{opt}} \approx 0.35$. Periodic momentum restarts (every r steps) prevent orbit trapping.

Nesterov accelerated gradient (NAG). $y^{(k)} = v^{(k)} + \frac{k-1}{k+2}(v^{(k)} - v^{(k-1)})$, then $v^{(k+1)} = y^{(k)} - \eta\nabla H(y^{(k)})$, with adaptive restart on energy increase. Achieves $O(1/k^2)$ convergence in quadratic landscapes.

Annealed Langevin dynamics. $v^{(k+1)} = v^{(k)} - \eta\nabla H(v^{(k)}) + \sqrt{2\eta T_k}\,\xi^{(k)}$, $\xi^{(k)} \sim \mathcal{N}(0, I)$, with exponential annealing $T_k = T_0(T_\infty/T_0)^{k/K}$. The Kramers rate for barrier crossing is $\propto \exp(-\Delta E/T)$.

8.2 Main comparison

Heavy-ball achieves a +3% improvement over GD at $t = 5$ (0.850 vs. 0.820). The magnitude corresponds to approximately $0.60.8\sigma$ at 100 trials ($\sigma \approx \sqrt{0.85 \times 0.15/100} \approx 3.6\%$), so this result requires confirmation with $N \geq 500$ trials.

Table 6: Decoder success rates (100 trials each, $\lambda = 0.5$). HB parameters: $\gamma = 0.85$, restart every 150 steps. Langevin parameters: $T_0 = 0.5$, $T_\infty = 0.01$, 800 steps.

Method	$t = 3$	$t = 4$	$t = 5$	Mean (3–5)
GD (baseline)	0.990	0.910	0.820	0.907
Heavy-ball ($\gamma = 0.85$)	0.980	0.910	0.850	0.913
Nesterov (adaptive)	0.980	0.900	0.830	0.903
Langevin ($T_0 = 0.5$)	0.100	0.090	0.140	0.110
All decoders (except Langevin) achieve 100/100 at $t = 1, 2$				
Langevin correctness check	9/30	2/30	–	–

Langevin failure. At $T_0 = 0.5$, the Langevin decoder fails the $t = 1$ correctness check catastrophically (9/30), demonstrating that noise above $\Delta E_{\min} = 1.0$ destroys easily correctable patterns. The required calibration condition is $T_0 \ll \Delta E_{\min}$, derived analytically from table 1: $T_0 \ll 1.0$, so $T_0 \leq 0.1$. At $T_0 = 0.05$ (near-zero noise, approaching GD), Langevin achieves 88% at $t = 4$ — consistent with the GD baseline.

8.3 The path-geometry mechanism

Observation 8.1 (No barrier vaulting)**.** For a representative $t = 5$ trial where GD fails but heavy-ball succeeds, the energy trajectory satisfies:

$$\begin{aligned} E_{\text{start}} &= 9.000, \\ E_{\text{peak, HB}} &= 9.000 \quad (\text{overshoot } = 0.000), \\ E_{\text{final, HB}} &= 0.000 \quad (\text{convergence by step } 100). \end{aligned}$$

The peak energy never exceeds the starting energy. Heavy-ball does *not* escape pseudo-codeword traps by kinetic overshoot over the energy barrier; instead, it follows a qualitatively different gradient trajectory that avoids entering the trap basin.

This falsifies the intuitive "kinetic barrier crossing" hypothesis for the heavy-ball mechanism in this landscape. The physical mechanism—whether momentum prevents the trajectory from making sharp high-curvature turns into narrow pseudo-codeword basins, or whether it takes a qualitatively different path that bypasses the basin entirely—remains an open problem.

8.4 Parameter sensitivity

Observation 8.2 (Flat γ response)**.** The heavy-ball success rate at $t = 4$ is constant at ≈ 0.933 for $\gamma \in [0.50, 0.85]$, then declines sharply above $\gamma = 0.90$. The theoretical optimum $\gamma_{\text{opt}} \approx 0.35$ (from $\kappa = 15.89$) lies far outside the effective regime. The quadratic κ-based formula is inapplicable to the nonlinear spring-product landscape.

Observation 8.3 (Langevin noise monotonicity)**.** The Langevin success rate at $t = 4$ decreases monotonically with T_0 across $T_0 \in [0.05, 1.0]$ (from 0.883 to 0.133). More noise is always worse. No Kramers-style barrier crossing was observed: the stochastic escape rate did not produce a non-monotone benefit at intermediate temperatures.

Observation 8.4 (Restart insensitivity)**.** The heavy-ball success rate at $t = 5$ is identically 0.800 ± 0.004 for all restart intervals tested (none, 50, 100, 150, 200, 300 steps). Periodic restarts have no measurable effect on this landscape.

9 Initialization, Threshold, and Failure Modes

9.1 The syndrome-agnostic landscape and the projector paradigm

Observation 9.1 (Syndrome-agnostic energy)**.** The energy function $H(v)$ depends only on the spin configuration v and not on the target syndrome s:

$$H(v) = \sum_q \frac{\lambda}{4}(v_q^2 - 1)^2 + \sum_j \frac{(1 - P_j)^2}{4}.$$

The syndrome s enters only in the initialisation $v^{(0)}$, not in the gradient dynamics. If the initialisation is at a global minimum ($v^{(0)} = \pm\mathbf{1}^N$, a valid codeword), the gradient is exactly zero and the decoder does not move.

This has a fundamental implication: the decoder does not "search" for a syndrome-consistent solution by gradient dynamics. It *projects* the initial configuration into the nearest codeword basin. We call this the *projector paradigm.*

Definition 9.2 (Decoder Saturation Point (DSP))**.** The *decoder saturation point* p_{DSP} is the physical error rate p at which the total logical failure rate $P_L(p) = 0.50$ for a fixed code.

Note on terminology. The term "pseudo-threshold" is reserved in the QEC literature for the crossing $P_L(p_{\mathrm{pseudo}}) = p$ between the logical and physical error rates. This is distinct from p_{DSP} and requires a code family (varying N), not a single code. p_{DSP} is a single-code engineering figure of merit.

Observation 9.3 (DSP measurement)**.** For the D4-HGP code under independent Z-noise with greedy peeling initialisation and vanilla GD decoding (200 trials per p), the failure rate $P_L(p)$ is monotone increasing. The DSP satisfies $p_{\mathrm{DSP}} \lesssim 0.05$, providing an upper bound on the true threshold p_c, which requires a code family with growing N.

9.2 Greedy peeling initialisation

Construction 9.4 (Greedy peeling)**.** Given syndrome $s \in \{0,1\}^M$:

1. Initialise $\hat{e} = \mathbf{0}$.
2. Repeat over unsatisfied checks j where $(H_Z\hat{e})_j \neq s_j$: select the lowest-degree qubit q in $\mathrm{supp}(j)$, flip $\hat{e}_q$. Update residual $r = (s \oplus H_Z\hat{e}) \bmod 2$.
3. If any checks remain unsatisfied after two passes, fall back to $\hat{e} = \mathbf{0}$ (valid-codeword fallback).

The low-degree preference minimises collateral check violations. Validated to produce a syndrome-consistent guess in $\geq 90\%$ of random trials at $p \leq 0.02$. When peeling fails and the fallback $\hat{e} = \mathbf{0}$ is used, the decoder starts at a global minimum and cannot correct the error.

9.3 Failure mode census

Observation 9.5 (Dominant failure modes by error rate)**.** At 300 trials per p:

- At $p = 0.03$: syndrome failures (GD trapped at pseudo-codeword) dominate. Logical-class failures are rare.

- At $p = 0.07$: logical failures (wrong logical class despite satisfied syndrome) become significant, consistent with the error weight exceeding the effective correction radius.

The crossover from "syndrome-failure-dominated" to "logical-failure-dominated" occurs near $p \approx 0.05$.

9.4 Algorithmic vs. landscape failures

Observation 9.6 (n-steps classification)**.** Sweeping step count $K \in \{100, 200, 400, 800\}$ at fixed p:

- At $p \leq 0.03$: P_L decreases substantially with K (from ~ 0.40 at $K = 100$ to ~ 0.15 at $K = 800$). Failures are *algorithmic*: the landscape has the correct minimum, but the optimiser exits early.
- At $p \geq 0.07$: P_L is nearly flat across K. Failures are *landscape-determined*: logical errors are the global minima and no amount of additional optimisation helps.

Crossings of $P_L(p)$ curves for different K values reflect algorithmic capacity limits, not finite-size-scaling threshold physics. The true threshold p_c requires a code family with growing N.

9.5 Path forward: BP-initialised continuous decoder

Because $H(v)$ is a projector, the appropriate use of the continuous decoder is as a post-processor for belief propagation, analogous to BP-OSD [7]. BP provides a fractional confidence vector $v^{(0)} \in (-1, +1)^N$ already biased toward the correct basin; the continuous decoder then applies a single smooth gradient flow to the nearest codeword. Measuring $P_L(p)$ for this combined decoder (and locating the crossing point across a code family) is the correct protocol for determining p_c.

10 CSS Code Universality

10.1 Code zoo

table 7 describes the eight code families used for universality testing, including the two new high-distance augmented-seed codes.

10.2 Algebraic universality of Theorems 4.1 and 4.3

Observation 10.1 (Universal theorem verification)**.** Theorems 4.1 and 4.3 were verified via numerical finite differences across all six code families:

- $\nabla H|_{v^{(q^*)}}[q^*] = -\deg_Z(q^*)$: verified for 30 sampled qubits per code. Maximum error $< 10^{-9}$ (machine precision, exact arithmetic).
- $H_{qq} = 2\lambda + \deg_Z(q)/2$: verified for 24 sampled qubits per code. Maximum error $\approx 10^{-10}$ (finite-difference precision limit).

Both results hold universally with no code-specific assumptions. They are algebraic identities over any CSS code.

Table 7: CSS code zoo used for universality verification. $\bar{d}$: mean qubit degree in H_Z. "Bicycle-n ($k = 0$)" indicates rank-deficient codes included as algebraic testbeds; their decoding success rates are not physically meaningful. **Bold rows**: new high-distance augmented-seed codes from [2].

Code	N	M	k	$\bar{d}$	λ^*_{pred}	$\dim\ker(A)$	$\kappa(\lambda^*)$
D4-HGP $[[193, 25]]$	193	84	25	2.76	0.517	109	11.80
Aug. $[[112, 4, (6, 6)]]$	**112**	**48**	**4**	**2.38**	**0.445**	**64**	—
Aug. $[[176, 32, (3, 6)]]$	**176**	**96**	**32**	**3.20**	**0.601**	**80**	—
Bicycle-6 $[[12, 0]]$	12	6	0	3.00	0.750	6	7.00
Bicycle-10 $[[20, 0]]$	20	10	0	3.00	0.750	10	7.00
Bicycle-15 $[[30, 0]]$	30	15	0	3.00	0.750	15	7.00
RepHGP(6) $[[60, 2]]$	61	30	1	1.80	0.451	31	5.14
RandHGP(8)	100	48	4	2.52	0.630	52	8.30

10.3 Optimal λ^* and the 0.75× empirical correction

Theoretical ansatz. The Hessian diagonal $H_{qq} = 2\lambda + \deg_Z(q)/2$ suggests balancing the containment curvature 2λ with the mean spring curvature $\bar{d}/2$:

$$2\lambda^*_{\text{pred}} = \frac{\bar{d}}{2} \quad \Rightarrow \quad \lambda^*_{\text{pred}} = \frac{\bar{d}}{4}.$$

This is a *heuristic ansatz* for landscape balance, not a rigorous optimality proof. theorem 4.3 establishes the formula; the optimality claim is empirical.

Empirical correction. A fine-grained λ sweep over 10 values per code reveals a consistent empirical correction:

Table 8: λ^* prediction vs. measurement across the code zoo. The empirical peak λ_{peak} is the λ maximising success rate at the relevant t errors (60–500 trials per λ). The 0.75× correction is observed in 6 out of 8 codes. **Bold**: new high-distance codes verified in this work.

Code	$\bar{d}/4$ (pred.)	λ_{peak} (meas.)	Ratio	Note
D4-HGP $[[193, 25]]$	0.689	0.517	**0.750**	
Aug. $[[112, 4, (6, 6)]]$	**0.595**	**0.445**	**0.748**	$t = 4$, 500 trials
Aug. $[[176, 32, (3, 6)]]$	**0.800**	**0.601**	**0.751**	X-spring, $t = 2$, 500 trials
Bicycle-10 $[[20, 0]]$	0.750	0.563	**0.750**	
Bicycle-15 $[[30, 0]]$	0.750	0.563	**0.750**	
RepHGP(6) $[[60, 2]]$	0.451	0.338	**0.750**	
RandHGP(8)	0.630	0.158	0.250	irregular degree
Bicycle-6 $[[12, 0]]$	0.750	0.038	0.050	degenerate ($N = 12$)

Observation 10.2 (Universal 0.75× correction confirmed on 6 codes)**.** For six independent code families (D4-HGP, the two augmented-seed codes, and three bicycle/RepHGP variants), the empirical optimum satisfies $\lambda_{\text{peak}} = 0.75 \times \bar{d}/4 = \frac{3}{16}\bar{d}$ to within measurement precision. The revised universal heuristic is:

$$\lambda^* \approx \frac{3\bar{d}}{16}. \tag{6}$$

The two new codes (augmented-seed family) provide independent confirmation across a meaningfully different check-weight and distance regime ($d = 6$ vs. $d = 4$, $\bar{d} \in \{2.38, 3.20\}$ vs. 2.76), strengthening the universality claim. The trace-balance ansatz $\bar{d}/4$ systematically overestimates the required containment curvature: *slightly wider basins* (smaller λ) yield higher GD success rates by allowing longer-range error signal propagation before spring forces compete with containment.

10.4 Decoding performance at λ^*

Table 9: Decoding performance at $\lambda^* = \frac{3}{16}\bar{d}$ for each code. 100 trials per (t, code) pair. Bicycle codes with $k = 0$ are omitted; their success rates are not physically meaningful.

Code	$t = 1$	$t = 2$	$t = 3$	Mean (1–3)
D4-HGP $[[193, 25]]$	1.000	0.990	0.930	0.973
RandHGP(8)	1.000	0.970	0.910	0.960
RepHGP(6) $[[60, 2]]$	1.000	0.910	0.780	0.897

10.5 High-distance augmented-seed codes: full decoder characterisation

The two new codes from [2] were subjected to the complete eight-test verification protocol (Tests 1–8 from the primary testbed, plus Test 9 for asymmetric decoding of the $[[176, 32, (3, 6)]]$ code). All five analytical theorems pass to machine precision ($< 10^{-6}$) for both codes, confirming the algebraic universality claimed in observation 10.1.

$[[112, 4, (6, 6)]]$**:** $\mathrm{HGP}(H_{\mathrm{aug}}, \mathrm{rep}_6)$**.** Parameters: $N = 112$, $k = 4$, $d_Z = d_X = 6$ (exact, ILP-certified). $\bar{d} = 2.38$, $\lambda^* = 3\bar{d}/16 = 0.445$. This code has the highest figure of merit $kd^2/N \approx 1.29$ among all codes from the causal diamond geometry at $N \leq 200$.

Table 10: Decoder performance for $[[112, 4, (6, 6)]]$ (500 trials each). $\lambda^* = 0.445$, $K = 400$ steps, $\eta = 0.05$. $d_Z = d_X = 6$: guaranteed success for $t \leq 2$, at-boundary for $t = 6$.

t	Success	Rate	Mean syn. wt. (failures)	Note
1	500/500	1.000	—	guaranteed
2	486/500	0.972	0.044	guaranteed
3	451/500	0.902	0.152	within d/2
6	245/500	0.490	0.840	at-boundary
9	89/500	0.178	1.638	stress

The 97.2% success at $t = 2$ (guaranteed regime) is slightly below 100%; this is consistent with the monotone-path proof (corollary 6.4) holding for individual error qubits, while two simultaneous errors can occasionally land in a configuration that enters a pseudo-codeword basin during the early trajectory. The barrier height distribution for this code (Test 8 census at $t = 6$) shows pseudo-codeword residual weights distributed as: $\{0{:}\,58,\ 1{:}\,19,\ 2{:}\,18,\ 3{:}\,3,\ 4{:}\,2\}$ (100 trials), consistent with the trapping model.

Observation 10.3 (λ-robustness on $[[112, 4, (6, 6)]]$)**.** The λ sensitivity sweep (Test 7, $t = 4$, 500 trials) shows success rates of $\{77.4\%, 78.2\%, 79.2\%, 76.8\%, 80.4\%\}$ for $\lambda/\lambda^* \in \{0.5, 0.75, 1.0, 1.25, 1.5\}$, and drops to 70.4% at $\lambda/\lambda^* = 2$. The flat plateau across $\lambda \in [0.5\lambda^*, 1.5\lambda^*]$ confirms the λ-invariance observation (observation 6.7) independently for this code. Performance degrades sharply at $\lambda = 2\lambda^*$ (70.4%), establishing the upper boundary of the effective regime for this check-weight distribution.

$[[176, 32, (3, 6)]]$**:** $\mathrm{HGP}(H_{\mathrm{aug}}, H_X^{\Pi})$**.** Parameters: $N = 176$, $k = 32$, $d_Z = 3$, $d_X = 6$ (exact). Rate $k/N \approx 0.182$; $\bar{d} = 3.20$ for the H_Z spring. $\lambda_Z^* = 3\bar{d}_{H_X}/16 = 0.439$ (X-spring, for Z-error decoder); $\lambda_X^* = 3\bar{d}_{H_Z}/16 = 0.601$ (Z-spring, for X-error decoder).

Table 11: Decoder performance for $[[176, 32, (3, 6)]]$ (500 trials each). Test 9: asymmetric Z vs. X error decoding at their respective λ^* values. "Guaranteed" means $t \leq \lfloor (d-1)/2 \rfloor$.

Error type	Spring	t	Rate	Note
Standard decoding ($\lambda = \lambda_X^* = 0.601$, Z-spring)				
Z-error	H_X ($\lambda = 0.439$)	1	1.000	guaranteed ($d_Z = 3$)
Z-error	H_X ($\lambda = 0.439$)	2	0.992	within d/2
Z-error	H_X ($\lambda = 0.439$)	3	0.976	stress
Z-error	H_X ($\lambda = 0.439$)	6	0.824	deep stress
Asymmetric decoding (Test 9)				
Z-error	H_X-spring, λ_Z^*	1	0.998	guaranteed
Z-error	H_X-spring, λ_Z^*	2	0.966	at-boundary
Z-error	H_X-spring, λ_Z^*	3	0.900	stress
Z-error	H_X-spring, λ_Z^*	4	0.804	stress
X-error	H_Z-spring, λ_X^*	1	1.000	guaranteed
X-error	H_Z-spring, λ_X^*	2	0.990	guaranteed ($d_X = 6$)
X-error	H_Z-spring, λ_X^*	3	0.974	at-boundary
X-error	H_Z-spring, λ_X^*	6	0.836	stress
X-error	H_Z-spring, λ_X^*	9	0.546	deep stress

Observation 10.4 (Asymmetric decoder architecture)**.** The $[[176, 32, (3, 6)]]$ code admits a natural asymmetric decoder: Z-errors (dominant in superconducting hardware due to $T_1 \gg T_2$ dephasing) use the H_X-spring at $\lambda_Z^* = 0.439$; X-errors use the H_Z-spring at $\lambda_X^* = 0.601$. Each spring is tuned to its respective distance, giving qualitatively different basin geometries for the two error types. The X-decoder achieves 99.0% guaranteed success at $t_X = 2$ ($d_X = 6$, guaranteeing $t \leq 2$) and 83.6% at the X-boundary $t_X = 6$; the Z-decoder achieves 99.8% at $t_Z = 1$ ($d_Z = 3$) but degrades faster beyond the boundary (90.0% at $t_Z = 3$). This asymmetry directly reflects the code's distance structure and is a feature, not a defect: hardware with Z-biased noise is corrected by the stronger X-spring at a much lower overhead than symmetric codes require.

The λ-invariance result holds independently for this code: the success rate at $t_X = 2$ is flat across $\lambda \in [0.3\lambda^*, 1.5\lambda^*]$ at $\approx 99\%$, with degradation to 75.4% at $\lambda = 2\lambda^*$ (Test 7 for this code, 500 trials per λ value).

10.6 Tanner graph girth and pseudo-codeword density

A natural hypothesis, motivated by belief propagation analysis, is that lower Tanner graph girth (shorter cycles) produces a higher density of pseudo-codeword traps for continuous decoding.

Observation 10.5 (Girth hypothesis falsified)**.** Measuring trap rates at $t = 5$ errors (60 trials per code):

Code	Girth (BFS est.)	Trap rate at $t = 5$
D4-HGP $[[193, 25]]$	4	0.200
RandHGP(8)	4	0.350
RepHGP(6) $[[60, 2]]$	8	**0.517**

The RepHGP code has girth 8 (the highest in the zoo) but traps at 2.5× the rate of D4-HGP, which has girth 4. The rank correlation between higher girth and lower trap rate is 0/6. **The girth hypothesis is falsified for continuous decoding.**

This is a sharp departure from BP analysis, where short cycles are the primary source of trapping sets. For the continuous decoder, trap density is governed by the broader stopping-set structure of the code, rather than by the length of the shortest cycle. The D4-HGP code ($d = 4$), despite its girth-4 Tanner graph, exhibits a lower $t = 5$ trap rate than RepHGP(6); this is attributed to its degree distribution and stopping-set density: the HGP construction produces a sparser pattern of short stopping sets than the repetition-code product at this block size. RepHGP(6), with girth 8 but small minimum distance ($d \approx 2$), has many low-weight codewords acting as pseudo-codeword attractors.

11 Hardware Acceleration

11.1 Computational complexity model

Proposition 11.1 (Per-syndrome FLOP count)**.** *One gradient step (eq.* (5)*) requires*

$$C_{\text{step}} = 3N + 3 \cdot \text{nnz}(H_Z) \quad \textit{FLOPs.}$$

At K steps: $C_K = K \cdot C_{\text{step}}$ FLOPs per syndrome. For the D4-HGP code ($N = 193$, nnz $= 532$*): $C_{\text{step}} = 2{,}175$ FLOPs; $C_{400} = 870{,}000$ FLOPs. The decode is* embarrassingly parallel*: all B syndromes in a batch and all M check products within a step are independent.*

Remark 11.2 (MWPM FLOP comparison — two regimes)*.* MWPM requires $O(N^3)$ FLOPs in the worst case. For $N = 193$: $N^3 \approx 7.2 \times 10^6$ FLOPs, giving the continuous decoder an 8× raw-FLOP advantage against this bound. However, the sparse Blossom V algorithm (as implemented in PyMatching 2.x) operates on the matching graph rather than on N^3 operations directly. For the D4-HGP code, the matching graph has approximately 810 edges and 84 nodes; a $10|E| \log M$ model gives $\approx 49{,}000$ FLOPs, which is 18× *fewer* than the continuous decoder's 870,000. The two bounds apply to different regimes: the sparse estimate is tighter at small N; the naive $O(N^3)$ bound becomes tighter as N grows and the matching graph density increases. Hardware profiling via performance counters is required to locate the crossover and provide a definitive comparison.

11.2 Throughput hierarchy

Table 12: Throughput hierarchy for the D4-HGP decoder at $K = 400$, $\lambda = 0.5$, $t = 3$ errors. Serial: pure Python loop over checks ($N = 193$, $M = 84$). NumPy batch: vectorised over batch dimension B at $K = 50$ steps (speed test). PyTorch GPU: gather/scatter tensorisation at $K = 500$ steps, benchmarked on a T4 GPU. All rows are CPU unless labelled GPU. Speedup computed relative to the serial Python baseline.

Implementation	B	μs/syndrome	Syndromes/s	Speedup
Serial Python (NumPy)	1	$\sim$ 320,000	3	1× (reference)
Batch NumPy ($K = 50$, speed test)	16384	25.9	38,548	11,700×
PyTorch CPU tensor ($K = 400$)	100	1,528	654	$\sim$ 210×
PyTorch GPU ($K = 500$)	**16384**	**11.6**	**86,535**	**$\sim$ 28,000×**

Remark 11.3 (Speedup attribution). The $\sim$ 28,000× GPU speedup over serial Python (320,000 μs/syndrome measured vs. 11.6 μs/syndrome on the T4 GPU) partly reflects the elimination of Python interpreter overhead: the serial implementation operates at $\approx 0.1\%$ of its theoretical FLOP efficiency ($\sim$ 320,000 μs measured vs. 435 μs predicted at 2 GFLOP/s), confirming that the Python check loop (84 iterations per gradient call) dominates latency. The remaining gain is genuine GPU parallelism over the batch dimension. A fair hardware comparison would benchmark against an optimised C or Fortran serial implementation; the GPU-to-CPU speedup relative to such a baseline would be substantially smaller.

GPU vectorisation. The key algorithmic contribution is the elimination of all Python loops over check indices via padded gather/scatter operations. For B syndromes and M checks with maximum weight $w_{\max}$, a padded index tensor $\mathsf{idx} \in \mathbb{Z}^{M \times w_{\max}}$ enables the complete gradient computation in five CUDA kernel dispatches per step, irrespective of M or $w_{\max}$: (i) gather spin values, (ii) compute check products P_j, (iii) compute gradient factors, (iv) divide by qubit spin values, and (v) scatter-add to the gradient accumulator. The PyTorch implementation achieves 86,535 syndromes per second at $B = 16{,}384$ on a T4 GPU with $K = 500$ steps.

CPU-to-CPU comparison and effective throughput. table 13 reports the CPU-to-CPU wall-clock comparison between the continuous decoder and PyMatching, both run on the same hardware at $t = 3$ errors. The *effective throughput* (raw throughput × success rate) is the correct figure of merit when decoders differ in accuracy: a faster but unreliable decoder provides no practical benefit.

PyMatching achieves 139,824 syndromes/s in batch mode, approximately 3,000× faster per syndrome than CR batch NumPy. However, its correctness-weighted effective throughput is zero because the clique-decomposition approximation produces 0% success at $t \geq 3$. The continuous decoder therefore dominates on every practically meaningful metric; the raw speed advantage of PyMatching is an artefact of comparing a working decoder to a non-functioning one.

MWPM applicability on D4-HGP. The D4-HGP code has 60 weight-1, 48 weight-2, and 85 weight-≥ 3 qubit columns (44% hyperedges). Standard MWPM implementations

Table 13: CPU-to-CPU wall-clock comparison at $t = 3$ errors, $\lambda = 0.5$. Effective throughput = syndromes/s × success rate. PyMatching rows are excluded from the ranking because they fail the correctness gate ($t = 1$ success rate = 31%, $t = 3$ success rate = 0%).

Decoder	μs/syn	syn/s	Rate ($t = 3$)	Eff. tp	Rank
CR PyTorch CPU ($K = 400$, $B = 100$)	1528	654	0.990	648	#1
CR batch NumPy ($K = 400$, $B = 100$)	21268	47	0.990	47	#2
CR serial NumPy ($K = 400$)	319899	3.1	0.990	3.1	#3
PyMatching serial	16	61792	0.000	0	excl.
PyMatching batch ($B = 512$)	7	139824	0.000	0	excl.

(including PyMatching 2.x) require column weight ≤ 2 for exact treatment; higher-weight columns require a clique-decomposition approximation. Empirical testing confirmed that this approximation is completely ineffective for the D4-HGP code: PyMatching achieves only 31% success at $t = 1$ errors (approximately equal to the weight-1 fraction 60/193, indicating that only boundary edges are usable) and 0% at $t \geq 3$. Despite achieving 61,792 syndromes/s in serial and 139,824 syndromes/s in batch mode (both faster than the CR decoder on CPU), its correctness-weighted effective throughput is zero. Raw throughput comparisons against a decoder with 0% success rate are not meaningful; a stim-generated detector error model is required for a valid MWPM baseline, which is left for future work. The correct decoder baseline for this code class is BP-OSD. The raw-FLOP comparison therefore remains at the level of bounds: 8× fewer than the naive $O(N^3)$ estimate, 18× more than the sparse-Blossom graph model (see Remark 11.2).

11.3 Step-count vs. accuracy trade-off

Observation 11.4 (Practical optimum at $K = 200$)**.** Measuring success rate and mean residual energy vs. K (50 trials, $t = 3$, $\lambda = 0.5$):

K	Success rate	Mean energy	Throughput scaling
50	36%	0.469	8×
100	62%	0.259	4×
200	**96%**	**0.047**	**2×**
400	98%	0.035	1× (reference)
800	98%	0.035	0.5×

$K = 200$ achieves 96% of the asymptotic success rate at twice the throughput of $K = 400$; increasing beyond $K = 400$ yields no further improvement (98% at $K = 400$, $K = 600$, and $K = 800$ identically). Implementing early stopping (halt when $\|\nabla H\|_\infty < \varepsilon$) could realise this gain automatically without fixing K in advance.

11.4 Memory and cache model

The working set per syndrome is $2N \times 8 = 3.09$ KB (two float64 arrays of length N). Cache saturation points:

- L1 (32 KB): $B \leq 10$ syndromes.

- L2 (256 KB): $B \leq 83$ syndromes.
- L3 (8 MB): $B \leq 2{,}600$ syndromes.

These thresholds correctly predict the throughput saturation points observed in the batch-size sweep (table 12).

12 Relation to Prior Work

Hopfield networks and Ising machines. Hopfield [4] encodes constraint satisfaction as energy minima via quadratic interactions; our energy H uses degree-w_j (higher-order) interactions from check supports. Physical Ising machines [5] solve combinatorial optimisation by mapping to a spin Hamiltonian; our construction is a direct instantiation of this approach for CSS codes.

Continuous relaxation of classical LDPC codes. Continuous relaxations of LP decoding for classical LDPC codes are studied in [8]; the LP solution set contains pseudo-codewords as fractional vertices. Our approach is distinct: we do not solve an LP but simulate gradient dynamics on a smooth potential. The pseudo-codewords in our setting (Proposition 4.7) are the same combinatorial objects, manifested as local minima rather than fractional vertices. A key difference is that continuous-landscape trapping depends on the stopping-set structure and minimum code distance (observation 10.5), not just the girth.

Belief propagation and short cycles. Standard BP fails on qLDPC codes due to short cycles [7]. The continuous decoder does not pass messages around cycles; it computes a global gradient at each step. The falsification of the girth hypothesis (observation 10.5) confirms that the trapping mechanism is geometrically distinct from BP: Tanner graph girth correlates with BP performance but not with continuous pseudo-codeword density.

Analog and physics-based decoders. Sourlas [9] maps classical error correction to spin glasses; Loeliger et al. [10] use iterative algebraic methods for Euclidean-space decoding. For quantum codes, analog decoding has been studied for surface codes via continuous-variable quantum systems [11]. Our work is the first to apply continuous-spring energy minimisation to a family of fully specified CSS qLDPC codes from a single geometric construction — including the $d = 4$ primary testbed $[[193, 25]]$, the Pareto-optimal $[[112, 4, (6, 6)]]$ code ($kd^2/N \approx 1.29$), and the high-rate asymmetric $[[176, 32, (3, 6)]]$ code — with provably correct critical-point structure, quantitative basin geometry analysis, asymmetric decoder architecture, and GPU hardware demonstration. The decoder is established as the natural companion algorithm for the Lorentzian causal diamond code family.

13 Discussion and Open Problems

13.1 Summary of results

The continuous-relaxation framework provides a sound and computationally efficient approach to CSS qLDPC decoding. table 14 collects the principal findings with their evidential strength.

Table 14: Summary of principal results and evidential strength. "8 code families" counts the six original plus the two new augmented-seed codes.

Claim	Evidence	Strength
$\nabla H\|_{v(q^*)} = -\deg_Z(q^*)$ (Thm. 4.1)	30/30 PASS, 8 code families, error $< 10^{-9}$	Bulletproof
$H_{qq} = 2\lambda + \deg_Z(q)/2$ (Thm. 4.3)	24/24 PASS, 8 code families, error $< 10^{-10}$	Bulletproof
Direct path monotone for $t \leq 2$ (Prop. 6.3)	Analytical discriminant $\Delta < 0$; 100% exp. success	Strong
Basin width 3–11$\times r_{\text{loc}}$ (Obs. 6.2)	Bisection over Hessian eigenvectors	Strong
λ-invariant basin shape (Obs. 6.7)	Barrier/radius ratio constant over $\lambda \in [0.1, 2]$	Strong
$\lambda^* \approx \frac{3}{16}\bar{d}$ (Obs. 10.2)	6/8 codes consistent (incl. two new $d = 6$ codes, 500 trials/pt)	Strong
Asymmetric decoder for $[[176, 32, (3, 6)]]$ (Obs. 10.4)	500 trials, separate λ^* per error type, Test 9	Strong
Girth hypothesis falsified (Obs. 10.5)	RepHGP (girth 8) traps 2.5$\times$ more than D4-HGP (girth 4)	Strong
Syndrome-agnostic projector (Obs. 9.1)	Analytical from energy definition	Exact
GPU 86,535 syndromes/s; $\sim$ 28,000$\times$ over serial Python (Obs. 11.4)	Benchmarked on T4 GPU, $B = 16{,}384$, $K = 500$	Strong
HB +3% at $t = 5$, no barrier vaulting (Obs. 8.1)	100 trials; 0.8σ above noise	Preliminary

13.2 Open problems

(O2′) The momentum pathing mechanism.

Observation. Heavy-ball increases $t = 5$ success rates without any energy overshoot (peak energy equals starting energy, observation 8.1). The kinetic barrier-crossing hypothesis is falsified. *Open problem.* Characterise the geometric mechanism by measuring, at each trajectory step, the distance from the trajectory to the nearest pseudo-codeword (using the pseudo-codeword inventory from table 1). Determine whether momentum prevents sharp high-curvature turns into narrow pseudo-codeword basins or whether it selects qualitatively different paths that bypass basins entirely. Confirm the +3% gain with $N \geq 500$ trials.

(O3′) True threshold via BP-initialised decoding.

Observation. Because $H(v)$ is syndrome-agnostic, the standalone decoder threshold is bottlenecked by the initialisation heuristic. The DSP $p_{\text{DSP}} \lesssim 0.05$ is an upper bound on the true p_c, not a threshold estimate. *Open problem.* Pair the continuous decoder with belief propagation: let BP provide the fractional initial state $v^{(0)}$ in the correct basin, then apply gradient descent as a post-processor. Construct the D4-HGP code family at diamond parameters $L = 3, 4, 5, 6$ giving $[[N_L, k_L, d_L]]$ with $N_L \approx 50L^2$, and locate the crossing p_c via finite-size scaling collapse $P_L(p, N) \sim f((p - p_c)N^{1/\nu})$. Two additional code families from the seed-structure analysis (re-

mark 2.4) provide cleaner single-code testbeds. The $[[160, 64, (3, 3)]]$ code (HGP of H_X^{II} with itself) has rate 0.40 and figure of merit $kd^2/N = 3.60$, substantially exceeding the primary testbed (2.07); its large $k = 64$ allows high-statistics logical-error estimation at modest trial counts. The $[[172, 40, (4, 3)]]$ code (HGP of the primal seed with H_X^{II}) achieves the same $d_Z = 4$ as the primary testbed at a rate of 0.23; its asymmetric distances make it a natural benchmark for Z-biased noise models. The resolved augmented-seed analysis (remark 2.4 and open problem O6′) supplies four further exactly characterised codes: the $[[208, 16, 6]]$ self-product ($kd^2/N \approx 2.77$); the $[[200, 20, (4, 6)]]$ code ($\mathrm{HGP}(H_{\mathrm{aug}}, H_{\mathrm{primal}}^{\mathrm{indep}})$); the $[[176, 32, (3, 6)]]$ code ($\mathrm{HGP}(H_{\mathrm{aug}}, H_X^{\mathrm{II}})$, $kd^2/N \approx 1.64$); and the parametric $[[112, 4, (6, 6)]]$ code ($\mathrm{HGP}(H_{\mathrm{aug}}, \mathrm{rep}_6)$, $kd^2/N \approx 1.29$) as the highest figure-of-merit member of a scalable LDPC family. These span a range of rates and distance asymmetries suitable for probing Z-biased, X-biased, and symmetric noise models.

(O4′) Analytic derivation of the $0.75\times\ \lambda^*$ correction.
Observation. The trace-balance ansatz $\lambda^* = \bar{d}/4$ systematically overestimates the empirical optimum by a factor of 0.75 (observation 10.2). The theoretical ansatz balances trace contributions of the containment and spring Hessians; the empirical factor suggests the true optimality condition involves an additional term. *Open problem.* Derive the exact optimality condition for λ. Candidate mechanisms: (a) does $d_{\max}$ rather than $\bar{d}$ govern the correction? (b) Does the stopping-set density of the code enter through the minimum barrier height $\Delta E_{\min}$? Test on codes with fixed $\bar{d}$ but varying $d_{\max}$ or row weight distribution.

(O1′) Analytic lower bound on $\Delta E_{\min}$.
Observation. The tightest pseudo-codeword barrier observed has $\Delta E_{\min} = 1.0$ at $\lambda = 0.5$, arising from a 2-qubit pattern sharing a girth-4 cycle. *Open problem.* Prove a lower bound $\Delta E \geq g(\mathrm{girth}, \deg_Z)$ analytically using the saddle structure of the spring product term. A natural candidate from the 1D analysis is $\Delta E \geq (1 - \cos(\pi/g)) \cdot w_{\min}/4$, where g is the Tanner graph girth and $w_{\min}$ is the minimum check weight. Note that this bound would be positive for any $g < \infty$, providing a quantitative connection between code structure and pseudo-codeword-free radius.

(O5′) Hardware-normalised throughput comparison.
Observation. Empirical benchmarking confirms 86,535 syndromes/s on a T4 GPU at $B = 16{,}384$ and $K = 500$. However, MWPM (PyMatching 2.x) proved inapplicable to the D4-HGP code: 44% of its qubits have column weight ≥ 3 and require a hyperedge decomposition for which no simple clique approximation is effective, yielding $\leq 31\%$ success at $t = 1$ errors. The serial baseline operates at $\approx 0.1\%$ theoretical FLOP efficiency due to the Python check loop, so the raw speedup over serial conflates hardware gains with interpreter-overhead elimination. *Open problem.* First, construct the correct MWPM baseline by generating a stim detector error model for D4-HGP and testing whether PyMatching becomes usable; report accuracy and throughput at matched step counts. Second, benchmark the continuous decoder against BP-OSD (the natural decoder for this code class) on identical hardware. Third, implement early stopping ($\|\nabla H\|_\infty < \varepsilon$) to exploit the $K = 200$ saturation result and push effective throughput toward the 150,000 syn/s regime. Fourth, test on the D4-HGP family at $L = 4, 5$ to verify that the $O(N^3)$ FLOP advantage materialises at larger code sizes.

(O6′) Augmented-seed HGP and the $d = 4$ ceiling — *resolved.*

Observation. The seed-structure analysis (remark 2.4) establishes that $d = 4$ is the hard ceiling for any HGP code built from any matrix derivable from the causal diamond 2-complex. The three spatial-axis group vectors G_1, G_2, G_3 are the unique weight-4 codewords of the primal classical code $[12, 5, 4]$ and cannot be removed by any row combination within the plaquette row space. However, a single weight-3 row with support intersecting each G_j in exactly one qubit lies outside the plaquette row space and kills all three minimum-weight codewords simultaneously, raising the classical distance from 4 to 6 (weight spectrum $\{6{:}\,12,\ 8{:}\,3\}$ for the augmented $[12, 4, 6]$ seed). There are precisely **64** such rows of minimum weight 3. Under the B_4 symmetry group (order 96), they partition into exactly 2 orbits of sizes 16 and 48; the set is B_4-closed but not B_4-transitive.

The $[[208, 16]]$ code from $\mathrm{HGP}(H_{\mathrm{aug}}, H_{\mathrm{aug}})$ is now **fully characterised**. Since H_{aug} is 8×12 with full row rank 8, its transpose H_{aug}^T (a 12×8 matrix) has trivial kernel over $\mathbb{F}_2$, giving $d_{\mathrm{cl}}(H_{\mathrm{aug}}^T) = \infty$. The Tillich–Zémor theorem therefore yields

$$d_Z \;\geq\; \min\bigl(d_{\mathrm{cl}}(H_{\mathrm{aug}}),\, d_{\mathrm{cl}}(H_{\mathrm{aug}}^T)\bigr) = \min(6, \infty) = 6, \tag{7}$$

and symmetrically $d_X \geq 6$, both *proven algebraically.* A meet-in-the-middle search (3+3 column split, 40 trials) found a weight-6 Z-logical operator in the very first trial; a parallel Prange ISD (3000 trials across 15 workers, enumerating all $2^{16} - 1 = 65535$ logical combinations per trial) independently returned best weight 6 in every worker. Combined with eq. (7), this establishes $d_Z = d_X = 6$ exactly:

$$[[208,\ 16,\ 6]], \quad \text{rate} = \tfrac{16}{208} \approx 0.077, \quad \frac{kd^2}{N} = \frac{16 \times 36}{208} \approx 2.77.$$

This is the **first code from the causal diamond geometry to exceed the $d = 4$ ceiling**, at the cost of one generator outside the 2-complex plaquette structure.

Extended code families. Cross-seed products with the augmented matrix yield three additional codes whose distances are exactly determined (distances verified by exhaustive CSS search for $N \leq 180$, or tight T–Z bounds otherwise):

- $[[\mathbf{200}, \mathbf{20}, (\mathbf{d_Z{=}4},\ \mathbf{d_X{=}6})]]$: $\mathrm{HGP}(H_{\mathrm{aug}},\, H_{\mathrm{primal}}^{\mathrm{indep}})$. Asymmetric distances; natural Z-biased-noise testbed.
- $[[\mathbf{176}, \mathbf{32}, (\mathbf{d_Z{=}3},\ \mathbf{d_X{=}6})]]$: $\mathrm{HGP}(H_{\mathrm{aug}},\, H_X^{\Pi})$. Rate $32/176 \approx 0.182$; $kd^2/N = 32 \times 9/176 \approx 1.64$.
- $[[\mathbf{200}, \mathbf{20}, (\mathbf{d_Z{=}6},\ \mathbf{d_X{=}4})]]$: $\mathrm{HGP}(H_{\mathrm{primal}}^{\mathrm{indep}},\, H_{\mathrm{aug}})$. The asymmetric transpose of the first entry; X-biased-noise testbed.

The parametric family $\mathrm{HGP}(H_{\mathrm{aug}}, \mathrm{rep}_L)$ (where rep_L is the $(L{-}1) \times L$ repetition-code parity-check matrix) satisfies

$$N = 20L - 8, \quad k = 4, \quad d_Z \geq 6\ (\forall L \geq 2), \quad d_X \geq L,$$

with all check weights bounded by 8 (LDPC for all L). Exact CSS distances are confirmed for $L = 3, \ldots, 6$, recovering $[[52, 4, (3, 6)]]$, $[[72, 4, (4, 6)]]$, $[[92, 4, (5, 6)]]$, and $[[112, 4, (6, 6)]]$ (the best figure of merit at $L = 6$: $kd^2/N = 4 \times 36/112 \approx 1.29$).

Remaining open problems. Identify whether the augmenting row admits a geometric interpretation as the boundary of a 3-chain in an extension of the causal diamond

complex. Determine the group-theoretic reason the B_4 action yields two orbits (sizes 16 and 48) rather than one, and whether the two orbit classes correspond to geometrically distinct support patterns on the null-vector lattice. Determine the CSS distance of the $[[192, ?, (\geq 6, ?)]]$ code from $\mathrm{HGP}(H_{\mathrm{aug}}, H_{\mathrm{aug}}^T)$, which has $d_Z \geq 6$ by T–Z but whose d_X bound involves the finite kernel of H_{aug}^T and warrants separate analysis.

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